

VPA IB DP CS — Case Study Preparation Guide

□ VPA IB DP Computer Science — HL Paper 3 Case Study Preparation Guide

Victoria Park Academy | IB DP Computer Science | HL Extension

A comprehensive guide to understanding, researching, and mastering the HL Paper 3 case study examination. This guide covers what the case study is, how to approach it, research strategies, note-taking methods, past themes, practice questions, time management, and common pitfalls.

What is the HL Paper 3 Case Study?

English: The IB DP Computer Science HL Paper 3 is a 1-hour examination based on a pre-released case study issued by the IB approximately 6 months before the exam. The case study describes a real-world scenario involving computer science concepts, and students must answer questions that test their understanding, analysis, and application of those concepts. The case study is worth 20% of the HL grade and is unique to HL students — SL students do not sit Paper 3.

中文: IB DP计算机科学HL Paper 3是一场基于IB在考试前约6个月发布的预发案例研究的1小时考试。案例研究描述了一个涉及计算机科学概念的真实场景，学生必须回答测试其理解、分析和应用这些概念的问题。案例研究占HL成绩的20%，是HL学生独有的——SL学生不参加Paper 3。

Aspect	Details
Duration	1 hour
Marks	30 marks
Weighting	20% of HL final grade
Format	Short-answer and extended-response questions based on the case study
Release Date	Approximately 6 months before the examination session (May/November)
Language	Available in English, French, and Spanish

How to Approach the Case Study

Phase 1: First Reading (Week 1)

- Goal: Understand the scenario, identify key stakeholders, and map the main computer science concepts.
- Read the entire case study without taking notes — get a holistic sense of the scenario
- Identify: What is the organisation? What problem are they trying to solve? What technologies are mentioned?
- Highlight or underline technical terms you don't fully understand
- Create a concept map showing relationships between entities, technologies, and goals

Phase 2: Deep Research (Weeks 2 – 8)

- Goal: Build deep knowledge of every concept mentioned in the case study.

- Research each technical term using academic sources (IB CS textbooks, academic papers, reputable tech publications)
- Understand not just what the technology is, but how it works, why it's used, and what its limitations are
- Connect each concept to the IB CS syllabus topics (Topic 1-7)
- Find real-world examples beyond the case study to deepen understanding

Phase 3: Synthesis & Application (Weeks 9 - 16)

- Goal: Apply knowledge to hypothetical scenarios and practice exam-style questions.
- Create practice questions based on the case study — "What if the organisation wanted to...?"
- Work through past Paper 3 questions from previous case studies to understand question style
- Form study groups to discuss and debate different aspects of the case study
- Write model answers and compare with mark schemes

Phase 4: Examination Preparation (Weeks 17 - 20)

- Goal: Refine exam technique, time management, and confidence.
- Timed practice under exam conditions (1 hour, no notes)
- Review all notes and condense to a single-page summary
- Focus on areas of weakness identified in practice
- Ensure you can explain every concept in the case study clearly and concisely

Research Strategies

Primary Sources

- IB CS Subject Guide: Ensure you understand how the case study connects to syllabus topics
- Official IB Case Study Document: Read every word carefully — questions are based on specific details
- IB CS Textbooks: Kosta Tsiprous, David Watson, and other IB-approved texts often have case study chapters
- Past Mark Schemes: Understand how marks are awarded for different types of responses

Secondary Sources

- Academic Journals: IEEE Xplore, ACM Digital Library, Google Scholar for peer-reviewed research
- Reputable Tech Publications: Ars Technica, Wired, MIT Technology Review, Nature Computing
- Organisation Websites: If the case study mentions specific companies or technologies, visit their official documentation
- IB CS Teacher Resources: Share notes and resources with your teacher and classmates

Research Methodology

Step	Action	Output
1. Keyword Extraction	Extract all technical terms, acronyms, and named technologies from the case study	Master keyword list (20 - 40 terms)

Step	Action	Output
2. Definition Research	Find precise definitions for each keyword from authoritative sources	Glossary with sources cited
3. Contextual Analysis	Understand how each technology is used in the case study scenario	Annotated case study document
4. Syllabus Mapping	Map each concept to the relevant IB CS syllabus topic (1–7)	Syllabus–topic cross-reference table
5. Extension Research	Research beyond the case study: alternatives, limitations, ethical issues, future developments	Extension notes with critical analysis
6. Question Generation	Write potential exam questions based on your research	Practice question bank (20+ questions)

Note-Taking Systems

The Cornell Method for Case Studies

Divide your page into three sections:

Cue Column (left, 25%): Key terms, questions, and syllabus references

Notes Column (right, 65%): Detailed notes, definitions, diagrams, and explanations

Summary (bottom, 10%): One-sentence summary of the page's content

Example for a case study about autonomous vehicles:

Cue Column	Notes Column
Lidar Topic 3 Sensor input	Lidar (Light Detection and Ranging) uses laser pulses to create 3D maps of surroundings. In autonomous vehicles, it provides high-resolution spatial data. Limitations: expensive, affected by weather, limited range. Alternative: camera-based computer vision (Tesla approach). Syllabus link: Topic 3 — hardware, sensors, and control systems.

Digital Note-Taking Tools

Notion / Obsidian: Link notes together, create concept maps, embed diagrams

OneNote: Handwritten annotations on PDF case study, audio notes from discussions

Anki: Flashcards for key terms and definitions with spaced repetition

Miro / Mermaid: Visual diagrams showing system architecture and data flow

Past Case Study Themes (2014 – 2024)

Year	Theme	Key Technologies	Syllabus Topics
2014	Smart Cities	IoT sensors, traffic management, energy grids	Topics 3, 5, 7
2015	Precision Agriculture	Drones, GPS, soil sensors, data analytics	Topics 3, 5, 7

Year	Theme	Key Technologies	Syllabus Topics
2016	Medical Expert Systems	AI diagnosis, knowledge bases, inference engines	Topics 3, 5, 7
2017	Autonomous Vehicles	Lidar, machine learning, sensor fusion, ethics	Topics 3, 5, 7
2018	Quantum Computing	Qubits, superposition, entanglement, cryptography	Topics 2, 3, 5, 7
2019	Blockchain & Distributed Ledgers	Hashing, consensus algorithms, smart contracts	Topics 2, 3, 5, 7
2020	Natural Language Processing	Tokenisation, neural networks, sentiment analysis, translation	Topics 3, 5, 7
2021	Machine Learning in Healthcare	Neural networks, training data, bias, diagnosis	Topics 3, 5, 7
2022	Edge Computing	IoT, latency, bandwidth, distributed processing, 5G	Topics 3, 5, 7
2023	Generative AI & Large Language Models	Transformers, training data, hallucinations, ethics	Topics 3, 5, 7
2024	Climate Modelling & Simulation	Supercomputing, parallel processing, data visualisation, uncertainty	Topics 2, 3, 5, 7

Pattern Recognition: Notice that most case studies fall under Topic 3 (Computational Thinking), Topic 5 (Abstract Data Structures), and Topic 7 (Control). Ensure you have strong foundational knowledge in these areas before diving into the case study specifics.

Practice Questions by Type

Type 1: Knowledge & Understanding (4 – 6 marks)

Example: "Explain how Lidar sensors contribute to the autonomous vehicle's navigation system described in the case study."

Approach:

Define the technology in your own words (1 mark)

Explain its specific function in the case study context (2 marks)

Describe how it interacts with other components (2 marks)

Mention a limitation or trade-off (1 mark)

Type 2: Analysis & Application (6 – 8 marks)

Example: "The case study mentions that the organisation is considering replacing Lidar with pure computer vision. Analyse the potential benefits and drawbacks of this decision."

Approach:

Identify 2–3 benefits with justification from the case study (3–4 marks)

Identify 2–3 drawbacks with justification (3–4 marks)

Use a balanced structure — don't just list, explain the impact

Type 3: Evaluation & Synthesis (8 – 10 marks)

Example: "Evaluate the ethical implications of using AI-driven diagnostic systems in the healthcare scenario described. Recommend and justify one policy to address your primary concern."

Approach:

Identify 3–4 ethical issues (privacy, bias, accountability, job displacement) (3–4 marks)

Analyse the severity and likelihood of each issue (2–3 marks)

Propose a specific, actionable policy (2 marks)

Justify why this policy addresses the primary concern (2–3 marks)

Time Management in the Exam

Time	Activity	Marks Target
0:00 – 0:05	Read all questions carefully; identify which questions you can answer confidently	—
0:05 – 0:20	Answer all short-answer questions (1–4 marks each) — these are typically knowledge-based and quick	12–15 marks
0:20 – 0:40	Answer medium questions (4–8 marks) — analysis and application questions	10–12 marks
0:40 – 0:55	Answer extended-response questions (8–10 marks) — evaluation and synthesis	8–10 marks
0:55 – 1:00	Review answers, check for completeness, add any missing details	—

☐ ☐ Time Trap Warning: Do not spend too long on the first question. Many students lose marks because they run out of time for the higher-mark questions at the end. If a question is taking too long, mark it and move on — return if time permits.

Common Pitfalls & How to Avoid Them

Pitfall	Why It Happens	How to Avoid
Regurgitating the case study	Students copy sentences from the case study without adding analysis	Always explain why something matters, how it works, or what the implications are
Generic answers	Students give textbook definitions without connecting to the case study	Start every answer with "In the context of [organisation/scenario]..."
Ignoring limitations	Students only describe benefits, ignoring drawbacks and trade-offs	For every technology mentioned, ask: "What could go wrong? What are the costs?"
Missing ethical dimensions	Students focus purely on technical aspects	Always consider: privacy, security, bias, accessibility, environmental impact, social impact
Poor time management	Students spend 20 minutes on a 4-mark question	Allocate time proportionally: 1 mark $\approx$ 2 minutes. Move on if stuck.

Pitfall	Why It Happens	How to Avoid
No real-world examples	Students only use the case study scenario	Bring in external examples to show depth: "Similarly, Tesla uses..." or "In contrast, Amazon's approach..."
Weak evaluation	Students list pros and cons without making a judgment	End evaluation questions with a clear recommendation: "Therefore, the best approach is... because..."
Neglecting the syllabus	Students research the topic but forget to link to IB CS concepts	Constantly ask: "Which syllabus topic is this? How would the IB want me to explain this?"

Exam Technique Checklist

- ☐ Read the case study at least 3 times before the exam
- ☐ Know every technical term by heart
- ☐ Practise writing answers under timed conditions
- ☐ Use the case study organisation's name in your answers
- ☐ Include diagrams where appropriate (system architecture, data flow, network topology)
- ☐ Define terms before using them
- ☐ Use structured paragraphs: Point → Evidence → Explanation → Link
- ☐ For 6+ mark questions, plan your answer before writing
- ☐ Leave 5 minutes at the end to review
- ☐ Write legibly — examiners cannot mark what they cannot read

Teacher & Student Resources

Recommended Resources:

IB DP Computer Science Subject Guide (latest version)

IB CS Case Study (released by IB for your examination session)

Kosta Tsiprous — IB Computer Science: Case Study Companion

David Watson & Helen Williams — IB Computer Science (Hodder Education)

IB CS Teacher Support Material (TSM) on MyIB

CS Academy (csacademy.com) for algorithm practice

Ars Technica, MIT Technology Review for current technology context

教师和学生资源推荐：IB DP计算机科学科目指南、IB发布的案例研究、Kosta Tsiprous案例研究 companion、David Watson & Helen Williams教材、MyIB上的教师支持材料、CS Academy算法练习、Ars Technica和MIT Technology Review获取最新技术背景。

Sample 20-Week Study Plan

Week	Focus	Activities
1	First Reading	Read case study; create concept map; list unknown terms

Week	Focus	Activities
2	Keyword Research	Research 5 key terms; write definitions; find sources
3	Keyword Research	Research 5 more terms; create flashcards
4	Syllabus Mapping	Map all terms to IB syllabus topics; identify gaps
5	Deep Dive 1	Focus on the primary technology/system in the case study
6	Deep Dive 2	Focus on secondary technologies and their interactions
7	Real-World Context	Research 3 real-world examples similar to the case study
8	Ethics & Limitations	Analyse ethical issues, limitations, and alternatives
9	Question Generation	Write 10 practice questions; attempt 5
10	Peer Discussion	Study group: discuss case study, debate alternatives
11	Past Paper Practice	Complete 1 past Paper 3; mark using mark scheme
12	Past Paper Practice	Complete 2nd past Paper 3; identify weak areas
13	Weak Area Focus	Targeted revision on topics with lowest scores
14	Mock Exam	Full timed mock under exam conditions
15	Mock Review	Analyse mock; create action plan
16	Extended Response Practice	Focus on 8–10 mark questions; write model answers
17	Condensed Notes	Create one-page summary of entire case study
18	Timed Practice	Complete 2 more past papers under timed conditions
19	Final Review	Review all notes, flashcards, and model answers
20	Exam Week	Light revision; confidence building; rest